158.337 Group Project Instructions: Part B

You will continue to work in groups* for this assignment. You do not need to register again but in case you change your group membership please let us know via emailing Indu (i.b.sofat@massey.ac.nz). Make sure you formally inform your Assignment Part A partner(s) as well. You will lose significant marks if you **do not work in a group without a very strong reason and permission**. You can use the group forum for this should you need to but do this early on; in fact, sort this first. Any other submission irregularities will also have a penalty.

Use the following instructions to guide you in completing **Four Sections (Part A-Part D**) of this project. You should turn the work in for grading by the due date **(Tuesday, 17th Oct 4.00 pm).**

Notes:

- *Working in groups doesn't mean you divide the sections and work. You should all work on each section and then compare your work and have a discussion around differences (different results, style of coding, getting stuck, etc.). Then put your best work in your report.

- Make sure you have altered the password of your (Group) Oracle a/c and keep it safe within your group. You do not want other people accessing your account and code. We have seen this in past that students have been able to access other accounts if the default passwords are not changed. Remember it is your responsibility to protect your work.

PART B

Queries (SQL/LINQ), Triggers, Procedures, Functions (PL/SQL) & MongoDB (NoSQL)

Download following two files provided in a zipped folder (partb.zip) under the Assignment heading from the Stream site:

- **Bookscript database file (Bookscript2023.sql) - for Sections A, B and C**.

The file provides code to create the required database in your Oracle account. Seven tables created are CUSTOMER, BOOKORDER, PUBLISHER, AUTHOR, BOOK, BOOKORDERITEM, and BOOKAUTHOR.

- **Collection file (NoSQL.txt) - for Section D.**

Use **script files** (use some text editor e.g., Notepad++, etc) for writing your assignment solution code. Name your files appropriately (e.g., SectionA.sql). Include query number (e.g., a) along with the given query question (e.g., Write a query that will list...) as part of the comment before writing the solution code (SQL / PL/SQL / LINQ) in these files. Annotate your code with suitable comments wherever necessary.

Use a **word file for your project report** (this can later be converted into a pdf should you wish to submit a pdf report). As mainly report will be used for grading so you will

need to transfer your final working code from the script files to this report as well. Here you also need to include the outputs of your code (e.g., query results, etc.). The outputs presented in the report should be well formatted (properly aligned, appropriate column names, etc.) for readability purposes. The script files will be used wherever required (i.e., we may not the run code for every question but may for the some to confirm that the code works correctly in differently simulated situations).

For PL/SQL (triggers, procedures, etc.) besides the creation code, the output produced from creation (e.g., procedure successfully created, etc.) you also need to provide the testing examples using some quality test data to prove that procedure/trigger works the way it is intended to work.

All (SQL/LINQ) queries must **only** be based on the information provided in the assignment question itself (e.g., Write a query that will list...). **Do not** use a different criterion to arrive at the expected equivalent query result set. Also, **do not** manipulate the query results by using **ROWID, ROWNUM, With Ties,** etc. to sort and get the first ranked /last record, etc. Some SQL may need use of subqueries, etc.

Please make sure the query results display the asked information along with the columns that are necessary to easily evaluate your query results. That doesn't mean using "*" to display everything – use your judgement and fulfil the query requirements. Your code should work correctly even when database is modified (e.g., added more records, deleted some, modified some values, etc.). Making some changes to the database and testing your code can take care of this.

Notes:

- For all questions asking you to display names, list first name and last name as a ***single column*** instead of separate columns.

- Remember to include all three - the **asked question** (as comment), the **code (SQL /PL SQL /LINQ/ MapReduce)** and the **output** (including any test cases, etc.) in your report. Some questions may have other additional requirements, so meet those as well.

Section A (SQL Queries) (20 marks)

Notes:

- For each of the questions (a)-(h) in Section A, write code using a **SINGLE** statement ONLY (i.e., you cannot write two separate select statements to arrive at the answer; however, select clause can be used more than once in a statement (There should only be one semi-colon in your code for each query.). Use the **given** criteria only.

- We will either give **full or zero mark** for most of the questions in this section so make sure your answer is fully correct by paying attention to all the stated requirements.

a. Write a query that lists both customer and author details (only their ids, first and last names). Provide suitable headings for the merged list. (1 mark)

b. Write a query that will list the books with the price difference (between retail and cost) of $20 or more. Display your results in the decreasing order of the price difference. (1.5 marks)

c. Write a query that will list books that have retail price of $35 or less and were published in any of the years 1999 or 2001. Display the year of publication values under the column titled as Publication Year. Show the query results in the increasing order of the publication year and decreasing retail price. (2 marks)

d. Write a query that will list publishers (with details such as name) and total number of books published by them. Include only those publishers that have 2 or more books published and display your output in the decreasing order of total number of publications. (2.5 marks)

e. Write a query to list books that have multiple authors. Also, display the number of authors who wrote the book. (2.5 marks)

f. Write a query that will list books from the '*computer*' category along with other details such as author(s), publisher, etc. List the book authors in a *single* row and the query should be able to accommodate all the casing-variations of the category values in the database. (3.5 marks)

g. Write a query that will list the customer who has ordered the greatest number of items (two copies of the same book will be counted as two items) to date. If there is more than one customer, then your query should list all those customers. (3.5 marks)

h. Write a query that will display the customer that referred the greatest number of customers. Again, like g) there can be more than one customer. (3.5 marks)

Section B (PL SQL) **(26 marks)**

Provide comprehensive testing examples for **all** PL/SQL (triggers, procedures, cursor, and function). You need to capture and show that you tested your code using good test cases. Consider including some exception handling if deemed necessary.

a. Think of some useful business rule or situation (based on the provided database) where it would be appropriate for your trigger to fire. However, do not write trigger to do something that could be done using some database design constraints (e.g., simple referential integrity checking, assign default values to attributes, or simply saying record is being inserted, or an attribute has a null value, etc.). Provide sensible and useful trigger and do not use the already provided or similar triggers (including tutorial triggers) for this question. Failing to meet these will result in a zero mark. There is also a discretionary mark for the degree of difficulty/relevance, etc. (1 out of the total 7). But **do not** confirm the trigger quality with staff before submission.

   Start first by clearly explaining the context, purpose (what it will do using business language) of your trigger. Then provide the PL SQL code and the results (using appropriate testing scenarios).

Trigger should be based on the tables already provided. Do not unnecessarily create too many and/or similar tables. Adding one or two tables may be fine – but justification is needed. Altering a table (adding a field) is fine.

Write one trigger (tell the trigger type). Display the successful creation and running of the trigger. Ensure that you also display the relevant tables before and after (results of the trigger) the trigger is fired. Remember to provide the purpose of your trigger (as stated in question b below), as this helps us to evaluate your work against the stated trigger requirements. (7 marks)

b. Write a trigger that does not allow the book retail price to be updated when the increase (in retail price) is over 25%. Provide test data and corresponding results to confirm that the trigger works. (4 marks)

c. Write a trigger that does not allow more than three author names to be associated with books under FITNESS category (e.g., if a Book is added, it should only allow up to 3 book authors to be recorded in BookAuthor table for '*fitness*' category books). Provide the appropriate test data and results. (4 marks)

d. Write a procedure to insert a new book record. The procedure should also automatically calculate the book retail value. This retail is calculated as 110.5% of the book cost price plus 4.5% of the average cost price of the existing books in the database. Provide rest of the attributes' values as input parameters. Execute your procedure to insert at least one book record. (4 marks)

e. Use a cursor construct to list book authors for the '*computer*' category books (along with their book title, cost, and retail). Present your work in a well readable report format (e.g., include headings, well-spaced expressions, some summary in the end). (4 marks)

f. Write a function to appropriately format money values (e.g., columns such as book cost, retail price uses a currency sign, commas if needed) when presented in some listing.
Call this function in a SQL statement for displaying books' costs and retail prices. (3 marks)

Section C (6 marks)

For (a) – (e) below, write queries using LINQ.

a. List books with the retail price $25 or over. (1 mark)
b. List books that have the word "HOW" in the book title. (1 mark)
c. List book categories, book titles and publisher names. (1 mark)
d. Display total number of books by each publisher in the order of publisher ID. (1.5 marks)
e. Display the number of books under each category in the order of category. (1.5 marks)

Section D (NoSQL) (8 marks)

Use the code provided in NoSQL.txt to create a collection of 15 rows called hobbits.

Now write **Map-Reduce code** to generate a report based on:

- gender-wise average weight

Rewrite the above MapReduce code using **Aggregation pipeline operators**.

Include both code and the report generated in your assignment report. Place the code in a script file as well.

Assignment Submission

Organise your final report (print friendly – but soft copy) to include all assignment sections' work. Make sure the report contents are also in the order of the laid assignment requirements (e.g., section-wise work).

There are three parts to this assignment submission:

- a report (word/pdf)
- scripts (code files – one for each section, clearly labelled) and
- database objects (in your Oracle Group account)

You must put your final report and code file(s) into a "single" compressed folder and submit via the Stream Assignment link.

Make sure there are not too many separate/unnecessary duplicated code files (e.g., all section A queries in a single file and not separate files). No duplicate group submissions (ONLY one per group).

Checklist for **final report:**
Readable format of SQL, PL/SQL, LINQ commands, and Map Reduce code. Remember to include the related **outputs** along with the code (not in appendix). For some questions, there are additional specific requirements (e.g., purpose of trigger, etc.).

Checklist for **submission under your Group Oracle a/c:**
All objects - database tables, triggers, procedures, etc.

Checklist for **Stream submission:**
Check your project work and make sure that all the scripts run without any errors. Also, check that appropriate names have been given to all file(s). Upload your report and code on Stream as a single compressed file on the Assignment link.

Notes:

Enter your names, student ids on the project marking sheet **(Appendix B). The project report should include this marking sheet (at the front i.e., 1st page). Make sure to provide the correct Oracle username in your report. Remember to include the script file(s). Make sure your work is well presented (i.e., easy to grade).**

Appendix B

158.337 Project Marking Sheet (Part B – 17.5% course mark)

(Attach this page to your project report BEFORE you turn it in.)

(Please make sure you provide all the necessary details)

Oracle Account: Group___

Group Member 1 - ID number, Name

_____

Group Member 2 - ID number, Name

_____

Group Member 3 - ID number, Name

Group Member 4 - ID number, Name

_____

(Grader's section, please do not write below this)

PART B: **_____/60marks**

Part B Comments:

Project Part B marks: /17.5

Total Project marks: /35